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Course: Microeconomics

Assignment #5

Due Date: 12/3/2024

Booth Honor Code Pledge

We pledge our honor that we have adhered to the Booth Honor Code during this assignment.

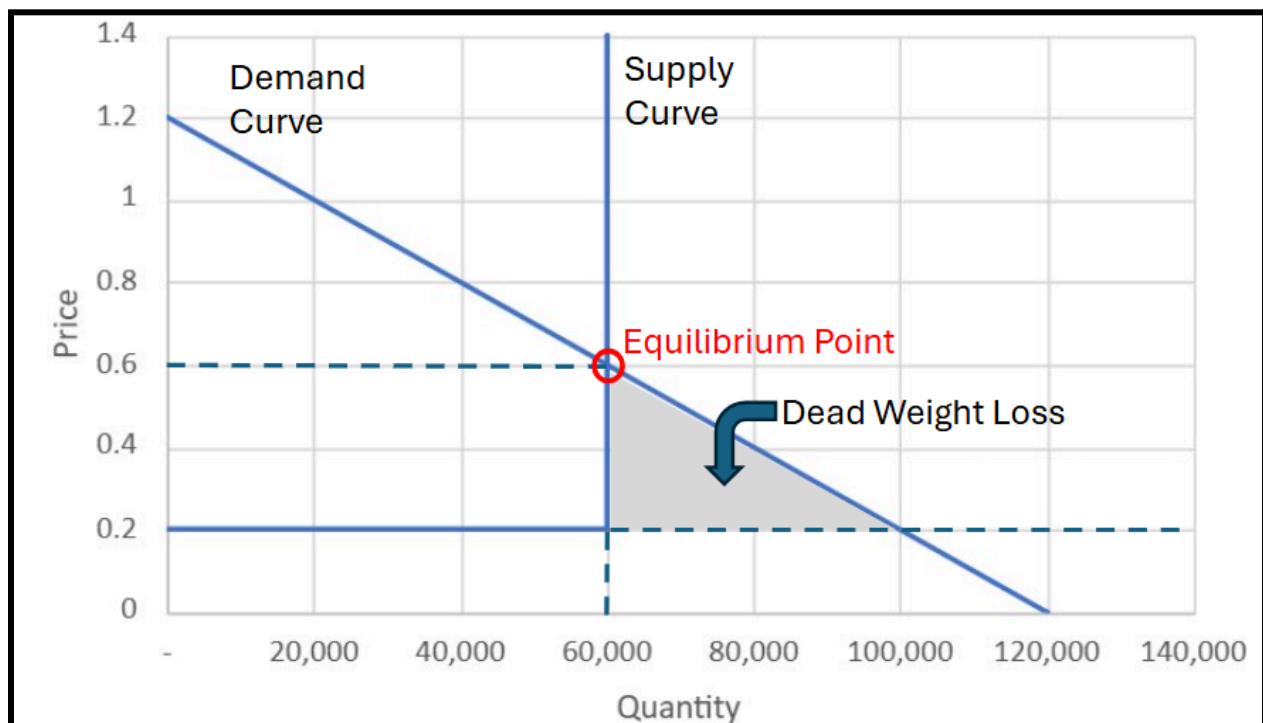
X Aaron Bennett Date 12/3/2024

Problem 1 Consider the taxicab example from class. Suppose that the market demand increases to

$$P(Q) = 1.2 - \frac{Q}{100,000},$$

but everything else remains unchanged. That is, there are 6 non-transferable taxi licenses and each taxi driver has the same opportunity cost function given by $C(q) = 0.20q$ with a capacity of 10,000 miles. What is (a) the equilibrium price, (b) the economic profit each firm earns from owning the license? (c) the deadweight loss of restricting the number of licenses to 6?

The graph below represents the situation described in the problem:



- a) The supply curve intersects the demand curve at 60K miles. At this quantity:

$$P = 1.2 - 60,000/100,000$$

$$P = 1.2 - 0.6$$

$$P = \$ 0.6 \text{ per mile}$$

- b) Each firm gains \$0.4 dollars of profit per mile driven (Price of \$0.6/mile minus the costs of \$0.2/mile) and drives 10K miles. Therefore:

$$\text{Profit} = (\$0.6/\text{mile} - \$0.2/\text{mile}) \times (10,000 \text{ miles})$$

$$\text{Profit} = (\$0.4/\text{mile}) \times (10,000 \text{ miles})$$

$$\text{Profit} = \$4,000 \text{ per month}$$

- c) The deadweight loss is the difference between the total surplus without the tax and the total surplus with the tax.

Without the license restriction

$$\text{Surplus}_{\text{Consumer}} = 1/2 \times (100,000 \text{ miles}) \times (\$1.2 - \$0.2) = \$50,000$$

$$\text{Surplus}_{\text{Supplier}} = \$0$$

$$\text{Surplus}_{\text{Total}} = \$50,000$$

With the restriction

$$\text{Surplus}_{\text{Consumer}} = 1/2 \times (60,000 \text{ miles}) \times (\$1.2 - \$0.6) = \$18,000$$

$$\text{Surplus}_{\text{Supplier}} = \$0.4 \times (60,000 \text{ miles}) = \$24,000$$

$$\text{Surplus}_{\text{Total}} = \$42,000$$

Calculating DWL

$$\text{DWL} = \$50,000 - \$42,000 = \$8,000$$

Problem 2 In the competitive market for DVDs, market demand is given by $Q^d = 500 - 20p$ and quantity supplied is given by $Q^s = 5p$, where quantity is given in millions of DVDs sold per year.

- (a). What is the market equilibrium price and quantity?
- (b). Suppose the government decided to tax DVD sellers by an amount of \$5 per DVD sold.
 (1) Under the tax, what after-tax price do consumers pay? (2). What after-tax price do sellers receive? (3). What is the new equilibrium quantity?
- (c). Describe, in words or with graphs, how does this tax generate a deadweight loss? That is, what is the source or story behind the deadweight loss?

- a) To solve this problem, we set $Q_d = Q_s$

$$500 - 20p = 5p$$

$$500 = 25p$$

$$p^* = 20$$

Plugging this p^* into the supply and demand curves, we get

$$Q^* = 100$$

- b) To solve this problem, we set $Q_d(P_b) = Q_s(P_s)$ and set $P_b - P_s = T$

$$500 - 20P_b = 5P_s$$

$$500 - 20P_b = 5(P_b - T) = 5(P_b - 5) = 5P_b - 25$$

$$525 = 25P_b$$

$$P_b^* = 21$$

$$P_s = P_b - T = 21 - 5 = 16$$

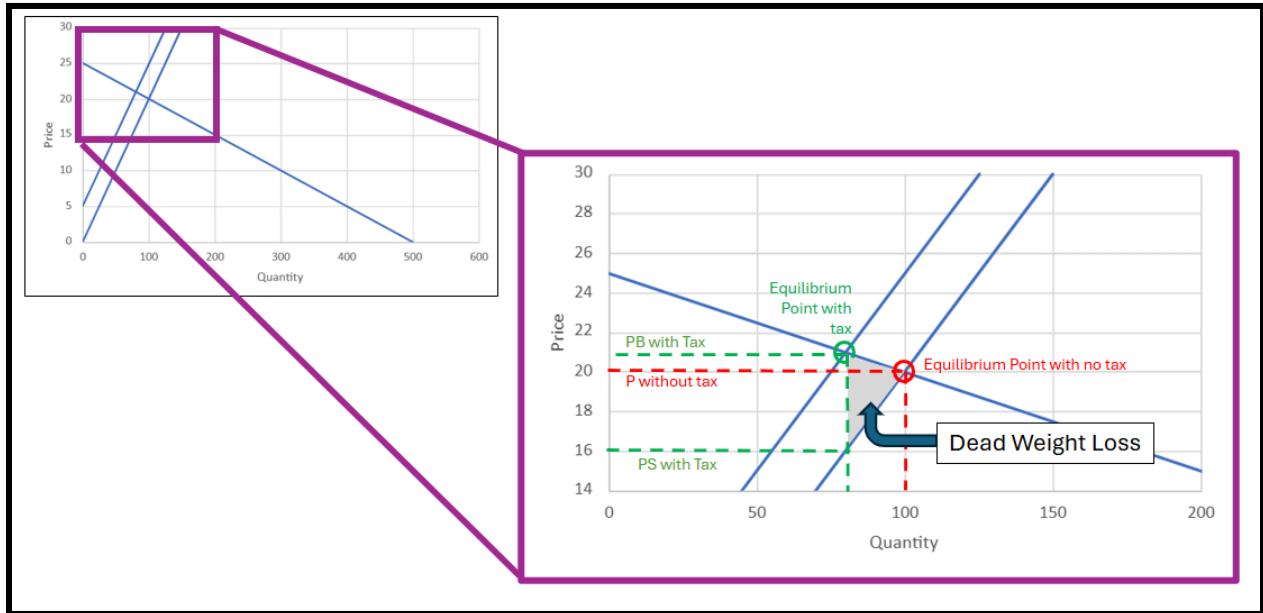
$$P_s^* = 16$$

$$Q = 5 \cdot 16 = 500 - 20 \cdot 21 = 80$$

$$Q^* = 80$$

- c) As shown in the figure on the next page (where quantity is shown in millions), the tax shifts the supply curve up (by the tax amount of \$5), which - among other things - reduces the quantity sold. This reduction represents a market inefficiency: the market would have sold 100 million DVDs without the tax, but now only sells 80 million DVDs. This decrease in quantity combined with the associated shifts in P_b and P_s results in the deadweight loss equivalent to the shaded grey area in the figure. The area of the shaded DWL region is: $0.5 \times (20 \text{ for change in quantity}) \times (\$5 \text{ tax}) = \$50$. Another way to view the DWL is that there are some sellers willing to sell at the price between \$16.01 to \$20, and some buyers willing to buy between the price of \$20.01 to \$21. However due to

the tax imposed, these groups of sellers and buyers cannot trade with each other. The inability to trade results in inefficiency, which is tabulated as the deadweight loss that represents the surpluses which would have benefited both the sellers and buyers (if the transactions would have occurred because the tax wasn't in place).



Problem 7 True, False, or Uncertain. [Explain your answer!] It has recently been argued that consumers see very little of gasoline taxes passed along into the price. Assume that the gasoline market is perfectly competitive and the taxes are levied per gallon. *This result implies that the supply of gasoline must be very elastic.*

Uncertain.

The percent of the tax that buyers or sellers pay depends on the relative elasticities of supply and demand (not just the elasticity of supply). Mathematically, the equation below shows how the percent of the tax burden carried by buyers depends on both the elasticities of supply and demand:

$$\frac{\Delta P_B}{Tax} = \frac{\epsilon^s}{|\epsilon^d| + \epsilon^s}$$

Because $\frac{\Delta P_B}{Tax}$ depends on both the elasticity of supply and demand, only knowing that the consumers see very little of the gas tax isn't enough information to make a definitive statement about just the elasticity of supply. At best, with the information we have we could only make a statement in relative terms about how the elasticity of supply compares to the elasticity of demand.